

CONTROL FLOW STATEMENTS

Control flow statements are divided into three categories:

- 1) Decision Making Statements
- 2) Iteration Statements(Loops)
- 3) Jump Statements

1) Decision Making Statements:

Decision making statements contain conditions.

If the condition is true then a set of statements are executed and if the condition is false then another set of statements are executed.

Decision making statements are also called selection statements.

- i) if Statement
- ii) if else Statement
- iii) if elif else Statement
- iv) Nested if Statement

i) if Statement:

Syntax:

if condition:

Statement1

Statement2

Example:

```
a=int(input("Enter any number: "))
```

```
if a>0:
```

```
    print("Positive Number")
```

ii) if else Statement:

Syntax:

if condition:

Statement1

Statement2

else:

Statement3

Statement4

Example:

```
a=int(input("Enter any number: "))
```

```
if a>0:
```

```
    print("Positive Number")
```

```
else:
```

```
    print("Negative Number")
```

iii) if elif else Statement:

Syntax:

if condition1:

Statement1

Statement2

elif condition2:

Statement3

Statement4

else:

Statement5

Example:

```
a=int(input("Enter any number: "))  
if a>0:  
    print("Positive Number")  
elif a<0:  
    print("Negative Number")  
else:  
    print("Zero")
```

iv) Nested if Statement:**Syntax:**

If condition1:

```
    if condition2:  
        Statement1  
        Statement2  
    else:  
        Statement3  
        Statement4
```

else:

```
    if condition3:  
        Statement5  
        Statement6  
    else:  
        Statement7  
        Statement8
```

Example:

```
a=int(input("Enter any number: "))
if a>0:
    if a%2==0:
        print("Even Number")
    else:
        print("Odd Number")
else:
    if a<0:
        print("Negative Number")
    else:
        print("Zero")
```

2) Iteration Statements(Loops):

A set of statements are executed repeatedly until the condition becomes false is called as iterative statement or loop.

- i) while loop
- ii) while loop with else block
- iii) for loop
- iv) for loop with else block
- v) Nested loops

i) while loop:**Syntax:**

```
while condition:
    Statement1
    Statement2
```

Example:

1) i=1

```
while i<=10:
```

```
    print(i)
```

```
    i+=1
```

2) i=5

```
while i>=1:
```

```
    print(i)
```

```
    i-=1
```

3) a=10

```
i=1
```

```
while i<=10:
```

```
    if a%i==0
```

```
        print(i)
```

```
    i+=1
```

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